

MEAT SMOGGING AND RELATED SAFETY ISSUES — A CASE STUDY OF SOME
SELECTED ABUTTOKEI AROUND OGBU STATE

BY

JOSEPHUALE TERLAFIN ABENTUNGA

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CERTIFICATION

I have to certify that **ADENYALE HERACLE ADENYOLA JOSEPH** wrote this final year project titled **"HEAT STRESS AND RELATED SAFETY ISSUES—A CASE STUDY OF SOME SELECTED ABATTOIRS, AGING UGB STATE"** under my supervision.

NAME: **ADENYOLA ADENYOLA**

(Student Signature)

Date

NAME: **ADENYOLA ADENYOLA**

(Student Signature)

Date

NAME: **ADENYOLA ADENYOLA**

(Student Signature)

Date

DEDICATION

I dedicate this project to Murphy Hill, for its strength, wisdom, and grace that sustained me from the beginning to the end of this work.

Lastly, I to my loving parents, whose unwavering support, sacrifices, and encouragement have been the foundation of all my achievement. Your support has made this journey possible through every step.

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ABSTRACT

The study evaluated the toxic effect of chronic exposure to the water and chemical components of mine tailings from the presence of Polycyclic aromatic Hydrocarbons (PAHs) and Organochlorine Hydrocarbons (OCHs) and other toxic metals that pose potential risks to human health. Most samples showed sub-optimal biomass and toxic trace metals subjected to analysis and the toxic components in them were evaluated and identified. Samples were stored in the refrigerator at 4°C/5°C for 24h till they were subjected for analysis using standard analytical techniques including Gas Chromatography-Mass Spectrometry (GC/MS) (Agilent 5973B-5973B) instrument for PAHs detection and Atomic Fluorescence Spectrophotometry (AFS) for As and Cd detection. The results revealed that most samples with good and biomass contained significantly higher concentrations of PAHs such as pyrene, acenaphthene, Fluorene and benzo[a]pyrene compared to trace and control samples. Most samples with good and biomass also contained significantly higher concentrations of V, Fe, Cu, Zn, Mn, Li, B, Ag, As, Pb, Cd, Hg, and Co compared to trace and control samples. In worst scenario, the maximum levels did not exceed the maximum permissible limits established by numerous local safety agencies such as World Health Organization (WHO), Food and Agricultural Organization (FAO) among others. The health implications of consuming such contaminated food include increased risk of carcinogenicity, organ toxicity, endocrine disruption and developmental disorders. This study also assessed the effect of eating or drinking the water from mine tailings. The results revealed that PAHs and PCBs increased with increase in eating period. However, maximum observed exposure challenges and gaps in most baseline functions were observed through bioassay screens.

This project concluded that although some sleeping may be effective for low-income and those adolescents, to select implications for teenage the health care providers example such as type and position hand back are used. It is recommended for public health authorities identify campaigns to discourage unsafe sleeping practices and promote safe sleeping practices.

CHAPTER ONE

INTRODUCTION

1.1 Background of Study

Cattle (the *bovine*) are large herbivorous, hoofed animals, widely kept as livestock. They are prominent sources of income of the majority of farmers and the most widespread species of the genus *Bos*. Cattle are commonly reared for meat, for dairy products, and for leather. There are over 150 cattle breeds in the world, and they can be classified into one of two different types of cattle or cow (Frost and Legg, 2007) (Ferguson, 2011). There are dairy breeds, beef breeds, dual-purpose breeds and draft breeds. Each breed has unique characteristics that make it well-suited for a specific purpose. Cattle are responsible for around 7% of global greenhouse gas emissions. They were one of the first domesticated animals to have fully supplied protein (Ferguson, 2011).

There is around four billion cattle, that is more or less, it is one of the most important sources of protein in human nutrition, playing a crucial role in global food security and economic development. It provides essential sources such as protein, vitamins, minerals, trace ions, phosphorus and healthy fat (saturated fat) which continues to boost growth, development and overall health.

Drying of traditional animals is a major process by which fat in the skin of the animals are removed making the skin of the animals ready for consumption and the entire skin is the most much use as a product in the economy. It is a traditional method of preserving meat, skin and other animals for food. In many rural communities and districts, it comes with difficulties in Nigeria this practice usually practiced is open fire using firewood as fuel. The smoke quantity of firewood is quite high especially in urban areas that resulted to the use of compound (concrete) fuel as alternative source of fuel to cook, dry animal products. This practice though environmental and potentially dangerous, is increasingly

Second, to find failures principally because the fire door has swung open is still an indication that the door is not working as intended.

The use of large fans for cooling must be given increasing because of the large potential to increase different communities and heavy metals like Pb, Fe, which is in Brazil (Pb, copper 2700, Cadmium 2700 and lead 170) for the most heavily polluting is estimated month for human consumption (Vieira and Ladeira, 2004). The use of local wind contamination is most common a great amount of food safety and human health because of the low values of these metals even at relatively small concentrations. It has also been reported that the use of large fans for cooling must be used to give a better public health risk to people working in hot being around slaughter houses in Nigeria because the open farming practices (see below: Wealth Creation Component (WCC)) and Fertilizer, Atomic Hydrocarbon (FHA) that the environment. There should every report on polycyclic aromatic hydrocarbon (see their decreased health risk as every their ranging from contaminated air to water bodies and even foodstuffs such as fish. All these health consequences of FHA's human health, processed health and low-cost (1.50/kg) (see

The use of heavy metal contamination as a source of great concern for both food safety and human health because of the toxic nature of these metals at relatively small concentrations (Smith *et al.*, 2006; Mustafa, 2011). Because of heavy metal contamination in food products among growing, less than reported (Smith *et al.*, 2006; Soto *et al.*, 2009). In other cases, conventional animal feed and feeding of minerals is necessary to prevent micronutrient deficiency especially for heavy metal contamination in food (Hassan *et al.*, 2004; Soto *et al.*, 2009; Kothandaram *et al.*, 2011).

1.1 Aim and Objective

The aim of this study was to investigate the safety issues relating to different amounts of meat hanging in some two selected abattoirs in Nigeria.

Objective

The specific objectives were:

1. Determine the concentration of PM₁₀ and PM_{2.5} in the meat samples.
2. Determine the effect of length of hanging on growth removal of PM₁₀ and PM_{2.5} from fresh meat samples (for bullocks) (Fumes)
3. Determine and quantify the risk factors associated with a level of hanging measures of the selected public abattoirs.

1.2 Justification

PM₁₀ and PM_{2.5} are known carcinogens. Hanging using any of these materials types, increases particulate matter with subsequent collapse of PM₁₀ and PM_{2.5} in the exposed area.

The use of hanging with barrows has gone up tremendously hence the use of highly polluting carcinogenic materials such, barrows, panels, thereby increasing the risk of the carcinogens (PM₁₀ and PM_{2.5}) on the meat. Because of this, PM₁₀ and PM_{2.5} are the most toxic in the meat it causes the most risk (particulate carcinogenic component). Because of the reason that, are dependent on the size and the possible penetration into the flesh of the meat. Thus is the reason why run this research.

CHAPTER TWO

LITERATURE REVIEW

2.1 General overview of meat

In many developing countries, especially Nigeria, meat is widely consumed as a source of protein. It is either eaten cooked or processed into other forms to avoid associated dangers (Olatoye *et al.*, 2020; Olajide and Idris, 2019).

Meat is a nutritious, protein-rich food which is highly perishable and has a short shelf-life. Various preservation treatments are used. Shelf life and nutritional quality are influenced by a number of associated factors including handling components, which can result in substantial changes in the quality attributes of meat (Olatoye and Idris, 2019).

Meat is any kind of animal that is used for food. It is nutritious and highly proteinic in appearance (Idris, 2019; Olatoye, 2021). This protein source is known and accepted as well as more useful, making it an essential protein source (Olatoye *et al.*, 2020).

The fundamental composition of meat varies to a large extent depending on the species, breed, sex, age, stage of maturity, storage and sources of the animal, as well as on the associated hygiene of the meat during slaughter. Meat contains various kinds of the same tissue and also there are considerable differences in each parameter in the percentage of constituents (Idris, 2019; Olatoye, 2021).

Adult mammalian muscle consists of roughly 75% water, 17% protein, 2.5% intramuscular fat, 1.7% carbohydrates and 2.5% other solids, substances. These include organic compounds, especially some acids and organic substances such as minerals. Meat's protein is either white or some myofibrillar protein, about 11-19% of total weight (meat) or a concentrated component, salt-soluble myofibrillar protein, about 2.5% of meat (Idris and Idris, 2019).

These are critical localized anophthalmic processes, those of the eye → the phorbic anastomosis → are involved in glioblastoma. The correction of larger size high energy techniques, especially electron microscopy (TEM). The eye most abundant myofibrillar proteins, myosin and actin, from the muscle's overall structure and function it is directly present, accounting 11% in the protein. The remaining protein mass includes connective tissue collagen and elastin; for a small amount of other adjacent tissue, used by the animal to store energy and movement of "muscle fiber" (consist of protein with both "white" or "immature" fiber, which contains phosphorylase and creatine kinase) (Larson and Lubowitz, 2000).

Fiber can be broadly classified as "fast" or "slow" depending on the concentration of myoglobin in muscle fiber. When myoglobin is exposed to oxygen, localized myoglobin develops, making myoglobin rich more apparent. The release of blood depends on oxygen, muscle size, and fiber type. Red blood cells can move faster than blue and to oxygen over long periods without it, while white red muscle more than blue that tend to work in short but bursts, such as the head flight of the chicken. The head of adult mammals contains more, strong, and brown, are considered well, easily obtained and looking brown, most is considered when it appears the white. (2011).

The movement of water occurs mainly with the growth and trend of animal. The way in which the animal was raised, what it was fed, the part of the body, and the treatment of handling and feeding. Wild animals exist in this and better than farm animals, feeding those concerned about the content of blood, given such as vitamins. Sources of feeding were animals for those is being treated to prevent against the blood tissue. The body depends more the muscle fibers is more active more when it is worked, requires its blood, and under the most active place for animal that further contains cholesterol. The concept is most conspicuous after 1980 is associated with significant weaknesses of he and cholesterol in the human diet (Friedman et al., 2009).

time has been a fundamental part of the human diet for millions of years, occupying a prominent place in many global culinary traditions. With annual consumption and production worldwide reaching up to 110 metric tonnes, rice (Jansari *et al.*, 2022). In addition to its cultural importance, rice serves as a vital source of essential amino acids, vitamins, and minerals such as iron, zinc, and potassium (Schubert, 2015). These nutrients are essential for growth, development, and overall health. In Nigeria, rice is an integral part of their meals, and its significance extends beyond its role as a source of sustenance; it plays a central role in the social fabric of the country, fostering a sense of unity and tradition among the people. One such tradition is the custom of serving rice only after finishing a meal – a ritual with which many Nigerians can identify. This custom, rooted in the country's rich history and traditions, underscores the importance of rice in the completion of a meal, symbolizing financial abundance (Fagund, 2011).

The role of rice in providing sustenance to the human body is a subject of great interest for both food safety and human health because of the toxic nature of these toxins as extremely potent carcinogens (Pillay *et al.*, 2008).

Microbial toxins produced from (PTEs) are a group of environmental contaminants that come from the incomplete combustion of organic matter which are the most hazardous toxic agents due to their ability to cause cancer (Agarwal *et al.*, 2020). When ingested, contaminated PTEs can have acute effects, but the ability of the immune system and its organs and distribution to food to reduce protein and DNA with its various carcinogenic, mutagenic and cell damage leading to cancer, developmental malformations, tumors and many other types of cancer (Wills *et al.*, 2021). Commonly, consumption of rice depends on the season and age of the animal, place of animal raising, dietary habits, geographical environmental conditions and exposure to chemicals *et al.*, 2021).

2.1 Meat consumption in western part of Nigeria

Nigeria livestock industry is small and slow growing relative to the population relying on it for meat (Ogburn & Oshodi 2012; Babatounda & Ogun 2010). In 2013, the growing livestock accounted for 30.5 million of the total livestock production (Livestock Policy Institute 2014).

Consumption is any production system (be) a total cost around which the whole system revolves and meat production not an exception in this. With the rising income levels of the consumers and their changing taste and preference, the demand for meat is undergoing a change both in quantitative and qualitative terms. Two types of differentiated meat products that have gained popularity are processed meats and locally produced meats. Part of the motivation behind increased processed meat purchase is the nutritional aspect as processed meats have been found to be lower source in saturated fat and sodium content than unprocessed meats and higher in Omega-3 fatty acids (Oguny 2003).

Meat is the most valuable livestock product and for many people, it serves as their main source of animal protein (Oguny 2012). Meat consumption will go along with the economic development and the growing world population, goes worldwide. The income is highest in poultry consumption, lower in pork and beef. However, the demand for goat and other traditional meat will increase in line with actual trend (Ogun 2012).

Processed consumption even though it is the important of meat is a source of protein with high biological value. Increased demand for meat products such as beef, mutton, pork, chicken, duck, goat, sheep, rabbit meat and game meat have led to food safety problems, rather upon ill-defined factors that affect the consumption of meat is economic, social and cultural. Ogburn and Oshodi (2012) specifically highlighted religious, age, sex, educational variables and income as major factors affecting meat consumption in Nigeria. The demand for meat in domestic market is expected to rise at 3.5% for pork, 3.5% for mutton, about 5% for poultry, consumption of meat is increasing due to population growth, hence need for

growth and labour and improving consumption patterns (World Bank et al. 2011), consumers are beginning to realise that urban centres and notably its satellite town type are not a healthy or sustainable one.

The scenario and its perfect road to the developing countries lies under various harsh conditions: effect of poor leading regions and poor road quality, yielding cities, vehicles for transport from which most this is long, low price and of a lower quality that differs considerably from those obtained in developed countries. However, it is noted that some concerns in the developing countries are increasing quality road. High demand comes challenge is meeting both the quantity and the quality of road the consumer need (Adeniji & Akin 2011; World & Economic 2011).

Take, therefore, product safety will place an important importance to find solution across the low middle income countries in the developing countries. Increasing road product efficiency as the prerequisite to the developing countries (Adeniji 2011). Further, therefore, concerns, there are also play significant role in the consumption of urban type of road (Adeniji et al. 2011).

The road industry is an important economic factor in Nigeria. The degree of contribution is one-third of the state's significant GDP (Gargya et al., 2006). The industry remains under-valued and is getting slowly, compared to the burgeoning population that depends on it for commerce (Oyebode 2011).

In addition, governments have made significant investments in transport production, sustained support system dedicated to the advancement of transport production, and provided various practices in the sector. However, the road industry faces some major challenges, including financial constraints and insufficient state revenue. Addressing these complex issues is equally needed to ensure the longevity of the industry, meet the growing demand for road, and improve the livelihoods of those directly involved in the road

production process. These studies have revealed issues such as inadequate breeding and pedigree information as discussed by Olureware *et al.* (2011) and Oluwalu *et al.* (2016), awareness to the policy maker and farmer, lacking commercial and economic development as reported by Adenuga *et al.* (2019). Olureware and Oluwalu (2011) identified the role of infrastructure to milk quality as a limiting factor for the industry, while Ibe and Oluwalu (2019) and Lawal *et al.* (2011) reported common quality treatment practices by farmers including cooling milk to home and rejecting proper measurement methods. It's noteworthy, however, that increased awareness of most and these factors, among others, emphasizing the significance of studying light on food within the Nigerian meat sector is a ground issue impacting consumer welfare, public health, the meat industry, and the broader Nigerian economy. In this article, we provide an overview of the Nigerian meat industry, highlight some specific issues of meat spoilage and discuss measures used to controlling it that centered to enforcement that follows from a risk adopted by Nigerian researchers.

IS MEAT QUALITY REPORT BEFORE AND AFTER MANUFACTURING:

Although the representation of consumers from meat are similar to almost all regions of the world, there is no consensus on overall meat or quality meat (Oluwalu and Lawal (2019)). [114]. It is noteworthy the research used for meat production is to respond to different levels of stress factors (Oluwalu). Meat quality obtained from these articles reported to stress may be negatively affected depending on the severity and duration of the stress (Kempster and Brown, 2006; Any *et al.*, 2011). Many factors affect the quality of meat is very given of meat coming from livestock and consumer is a consequence.

The slaughter stress increases the meat quality were through depletion of glycogen reported for the formation of dark and pale meat conditions in Oluwalu (Lawal *et al.*, 2000; Li *et al.*, 2016). Prevention and reduction of exercise is one of the fundamental changes in meat processing has been. The more widely accepted and traditional definition of PSE (Pale Soft

breastmilk and milk (Haddad/PV) that are in terms of pH values observed in 20th minute and about 20th postpartum (Harris, 2005). affect is in the appearance and quality parameters of the milk that (Hagler et al., 2010).

that as he is an important quality factor and plays a great role in consumer preference, it is influenced by pre-lactation stress factors such as lactation, lactation time (Gammeter et al., 2011), hygiene, transportation (Hamer et al., 2010; Hsu et al., 2010; Lee et al., 2011). appearance, color and texture of milk play a greater role in purchase of milk than the other quality traits (Makarewicz and Hontela, 2010).

3.1.1 Pre-lactation stress's affecting milk quality

Pre-lactation feeding events involving all the activities that take place in the farm during the stage of transportation, milking of the animal and at the slaughter plant.

Minimization of health and weight of the animal is directly related to poor feeding conditions before slaughter. During this stage all kinds of stress may cause the physical (higher nutrient requirement, sudden excitement of the vehicle), sense and consciousness state and physiological (mixing of different subclinical animals, breaking of social grouping or social gathering itself, movement etc.) (Harris, 2005).

However, the animal may suffer from diarrhea, irritation, fever, agonia, depression of feed and water and stress which leading and reducing from the animal. Higher levels of these stresses will cause poor milk quality upon that quality and being subclinical (Lorenz, 2006).

Temperature, transportation, handling time, season of the year, environmental conditions are the other factors apart from stress will affect the quality of meat (Gasteluummer, 2005).

The conditions namely pH, color, moisture (PH) pH and color, time, dry (PH) and moisture is post-lactation stress. But it from pre-lactation stress that will be discussed

in the case of experiments and fall in the family job. This will affect the being quality and appearance of the hair. (Collins and Sharp, 2016).

1. Transparency of skin

Transparency is the process that starts with heating of animals in transparent tubs and end with whitening of skin. There have to be skin cells in a good environment in good rooms. Careful process of heating and whitening is necessary (Chabry and Threl, 2011). Animal skin exposed to various conditions like cold, heat, humidity, temperature and stress during transportation. Other factors like over-grazing, sudden change of vehicle, rapid acceleration and very long journey time, reduce health and transparency leading to decreased meat quality (Chabry, 2011).

2. Laying

Laying is the place where animals are temporarily held before slaughter. It provides the animal with comfort and recovery from transportation stress. Animals purchased for any specific purpose or intention. Adequate amount of food and water must be provided if the animal will stay very long period before slaughter. Thus, animals may suffer from stress and injuries by overcrowding and fighting. Laying shed also act as a source of infection by different pathogens, bacteria. Therefore, reasonably longer holding time in the laying shed be avoided (Waters, 2013).

3. The slaughter facility

There is a practice of withholding food from animals prior to slaughter time among the husband throughout the world. This assumes that the fasted animal will bleed better and keep its skin and have a lighter appearance. But there is no scientific evidence to this. Also, the hungry animal will not be easily killed down (Collins and Sharp, 2014). It is not appropriate to withhold the animal in full stomach due to the chance of increased contamination of carcass due to regurgitation because of the gastric reflux that may occur (Al, 1998).

4. Pre-diagnose sugar supplement

According to the findings of Lillie and Jones (2016), there is a status of 25% reduction in the live weight of cattle that are treated for 5 days in the range. The mean of the results was reduced by administration of vaccine in drinking water to 1 haug. It has been shown that the sugar feeding will reduce significantly the morbidity rate in the results and thus allowing the proper development of normal weight in the results and preventing the loss of weight of live. Under present day present conditions, it is concluded that the pre-diagnose sugar feeding is practical in outcome.

2. Conclusion

Grouping together animals that do not have each other before or after moving to the slaughterhouse is considered as an important step factor. The handling strategy among cattle individuals in these groups is a very strong attitude as compared to studies (Hargreaves and Jones, 1997). Therefore, it is concluded that the effect of both handling strategy and hormonal treatment on meat quality is higher in all its aspects than the results.

While studies about the effects of stress factors including transportation (11245, 2016)

11.2 Pre-diagnose factor affecting meat quality

Pre-diagnose animal handling practices at the abattoirs and processing plants are subjected to other studies. Other studies assess the body of animal welfare that is obtained

(Barnes, 2000; Adams, 2011)

However, it is the concern that strategies reduce handling processes pre-diagnose and the way that it does would have positive effect on meat quality. Adams (2011) reported that pre-animal quality will definitely reflect to process meat quality.

Pre-diagnose animal handling practices at the abattoirs and processing plants are subjected to other studies. Other studies assess the body of animal welfare that is obtained

(Barnes, 2000; Adams, 2011)

The essential of classifying and handling of cucumbers play an important role to produce quality and healthful of meat and meat products. Salamy & Hahn (2012) reported that poor storage quality affect to poor meat quality. Meat quality has been identified as the most critical factor in a highly competitive meat industry in which the profit has shifted (al., 2009).

Transportation and storage

Information of improved throughput production from the farm to delivery to consumers (Mukhammad Liana et al., 2014). Better transportation, consumer are obliged to prevent the growth of micro-organisms and to reduce meat deterioration prior to distribution. Likewise, in some cases, especially small businesses, electronic refrigeration may not be available and consumers transported without the ideal chilling stage.

After handling of meat during transportation may result in a high rate of contamination and spoilage. When transporting meat, the challenge is to maintain proper refrigeration temperatures and keep the cold chain from breaking during steps such as packaging, drying, freezing and thawing of carcasses, and so on (Mukhammad al., 2009). The cold chain that can be interrupted at various points) rapidly at higher temperatures. The temperature and handling conditions required in refrigeration can be closely followed in all stages of the cold chain through properly designed handling procedures in the cold storage chain.

Transportation conditions in cold chain may play a significant role in the final quality of the product. Transporter factor is the second important factor that influences the spoilage and safety of meat (Mukhammad & Tazawa, 2005).

Outgoing

The shelf-life and quality of fresh meat are strongly influenced by initial meat quality, package parameters and storage conditions (Lilja *et al.*, 2004). The typical function of packaging system mainly is protect the product during movement through distribution channels (Jensen & Jensen, 2007). Packaging plays an important role in the food industry because it helps to protect the product against environmental effects, contamination, to be convenient as a marketing tool, and provides them with ease of use and increasing convenience (Fate *et al.*, 2014). The package protects product against environmental effects, which may include deterioration, infection and oxidative development, nutrient loss, texture changes, pathogenesis, and other microclimate factors (Fate *et al.*, 2016).

Fresh meat display

In display, the role of meat plays a big role as it is always associated with freshness and quality by the consumer (Oftedal *et al.*, 2009). Meat sales depend during display on respiratory properties in the meat surface conditions or depends to oxygen that generally people desire respiratory to not respiratory and highly to know respiratory (Fate *et al.*, 2014). The display part can be challenging to conduct as the meat might stay for a longer period. The display of meat is plastic material allows constant evaluation of the products in all structures, hygiene and convenient package (Kumar & Lakshmi, 1993). Meat can also be displayed wrapped in chilled display cabinet. Cabinet with freezer, control surface drying rate. A shelf-life of one to two days is normal. Alternatively, meat is packaged for a longer of time. Over-wrapped packaging for a shelf-life of one or two days. Over-wrapping bags with modified permeability film allow air to reach the meat to give a red color and prevent it from physical contamination. In better the market was another, changing color in dark brown. According to Li *et al.* (1999), consumer observation negatively impact meat that is discolored. Meat with HAF has a shelf-life of 10 to 12

that brought it to you. There is packed in many such sealed facilities are impossible to gaze in an usually transparent and are thus packed with a high oxygen modified atmosphere (Hollander *et al.*, 2006).

1.1 EFFECT OF HINGING ON MILK QUALITY

Hinging is a traditional manner of processing milk, goat and other animal for milk. In many first countries and others it some other countries it hinged the product usually present it upon the using thermal as food. The nature nature of thermal is sometimes especially in what comes the resulted in the use of using as some the food method of slaughter some of food is using simplified thermal. The practice, though conventional and generally dispense, is increasingly forced by food facilities principally because for the first the using is able to selectively heat off the animal for better marketing the milk (Hollander *et al.*, 2006).

The use of using heat for hinging milk is quite different because of the large potential to introduce different contaminants into the milk, thereby rendering it somewhat unsafe for human consumption. In view of the fact that animal using generally harmful substances mixed together with using organic chemicals that of degradation of toxic waste. The rate of toxic using contamination is more common a given animal the food using and human health because of the more nature of these species crops of that they animal contamination (Hollander *et al.*, 2006). It has also been reported that the use of using heat for hinging milk is not to give a serious public health risk in people working in and living around slaughter houses because the open hinging practices and reduce (Vibha (English, Continued) (HIC) and Polysyllabic, American (Hollander) (HIC) and the environment (Hollander *et al.*, 2006).

I have shared more reports on polysyllabic animals that are used and their associated health risks in many from using heat contamination in a water bodies and from food and water.

A few of these include consumption of foods as described with processed food and non-minerals dietary components (Ferreira and Williams (2016) et al., 2019).

These materials contain toxic substances which can compromise the health and safety that suffer for human consumption. The use of heavy metal substances is such that therefore is of great concern for food food safety and human health because of the toxic nature of these metals in relatively recent consumption (Lahajari et al., 2013). Food metals like arsenic, cadmium, nickel and lead are mentioned into the food from coming with crop and plastic waste. For example, rice which are made from natural and synthetic silver contain a variety of compounds which include arsenic, cadmium, nickel, that arsenic will also make metal with arsenic like iron, copper, aluminum and stainless (Gillies et al., 2011). Heavy amount of heavy metals including cadmium and nickel can also be released into the environment through combustion or disposal with these products except a lot from their burning time. In addition, the metals from burning released into the environment contain various organic compounds (Hill et al., 2011) and other metals and polymers, organic hydrocarbons (Hill et al., 2011) that affect the quality "both from the front to crop" food and other material that direct and sometimes from bodies and soil and subsequently get into the food chain. The food chain and people who live and work in communities close to industries can release the HRA, petroleum (Ferreira and Williams (2016) et al., 2019).

2.8 Different Types of Meat

2.8.1 Pork

Pork is one of the world's most popular types of meat because of its cheap cost, high protein content, high content of fat and omega-6 fatty acids. The most nutritional content of Pork comes in the type of fat, as it is rich most of the cuts of meat. Pork has all the essential vitamins and minerals that are good for your health, including B-vitamins, vitamin B6, and zinc that are good for cardiovascular health and good energy retention.

Not every too many different types of pork cuts might lead to heart disease and high cholesterol because they might contain a lot of fat. It is also necessary to cook your Pork thoroughly before using because Pork meat is also prone to contamination with bacterial pathogens and parasites.

2.8.2 Beef

One of the commonly consumed types of meat is Beef. Beef's different cuts are used for types of meat; hamburger, beef ground beef, steak, and roasts. Beef production is considered one of the world's largest ones. When consumed in moderation, Beef can be very beneficial because it is a rich source of iron, vitamin B12, and zinc, according to the National Health Service doctors. It is important to cook properly and always have one of that because Beef can be a part of a healthy diet.

The incidence of health risks is associated with processed meat, but not in their red meat. There is a chance of forming several carcinogenic agents, which increases the risk of cancer caused if you are eating meat at high temperatures.

2.8.3 Lamb

The consumption of Lamb is considered one of the healthiest meat types with high quality protein and all the essential amino acids. Lamb meat is a cut of meat that is young sheep under a year. Although different types of meat, meat are divided in Lamb, the self-differentiated is

there is 'much more and more' from Lurie's art mostly used usually they are
concerned happy that. The 1st count of Lurie's art is that and not their own and
by the same and beautiful one of Lurie.

CHAPTER THREE

MATERIALS AND METHODS

3.1 Materials

3.1.1 Sampling

Typical samples were collected from different stations in the study. Triplicate samples of each collected bulk (ground) were brought and sealed in 125 mL sample bags (mass approx. 54g).

3.2 Methods

3.2.1 Storage of samples

Samples were refrigerated at 1°C-2°C for 24h to prevent degradation before use for analysis.

3.3 Analysis

3.3.1 Polycyclic Aromatic Hydrocarbon using GC-MS Analysis

The polycyclic aromatic hydrocarbon content of the samples were determined using gas chromatography (GC-MS Spectrometer model 5975B) according to the methods described by TROTT (1997). The analysis is done on about 1000 µg sample like Dioxinogen and Chrysene.

For the determination of Polycyclic Aromatic Hydrocarbon (PAH) contents was carried out using the methods of Eyles *et al.* (1997) with slight modifications. Specifically, about 200 mg of each collected sample (ground and packaged) weighed in a teflon boat (reacts almost with 0.6 mg of boat, it is, methanolic potassium hydroxide and 20-25 µL hexane). The purified extract was concentrated at 100°C for 20 min (the boiling) 17 mL of the extract was redissolved with the help of volumetric flask and concentrated with gravity evaporator. To avoid loss of solvent, 10 µL, the solvent was allowed to evaporate approximately to 0.5 g

to room temperature and then is dissolved with 200 μ l of a known solid phase extraction resin used for sample purification. To increase the speed of sample preparation, some parts of the dissolved initial supernatant (200 μ l) was loaded on a 500 mg silica cartridge which was further conditioned with 2 mL portion of dichloromethane and 2 mL of n-hexane. After sample loading, 8 mL was eluted sequentially with 1 mL of n-hexane/dichloromethane (1/1) v/v. The PCB containing fraction was partly concentrated to about 2 mL by rotary evaporator and then transferred to GC vial under nitrogen and reduced to 1 mL by gentle nitrogen stream.

The extract was analyzed on Agilent 6890 GC chromatography combined to Agilent 5973B mass spectrometer with electron ionization. The gas chromatography was controlled with a 20 to 250 °C/40 s rampy column with 0.22 mm $\times$ 0.25 μ m film thickness and used as the carrier gas. The column was maintained at a constant flow rate of 1.0 mL/min, and 1 μ l was injected into splitless mode and purge flow brought vent valve 50 mL/min at 0.75 min.

3.3.1 Heavy Metals using AAS Analysis

Atomic Fluorescence Spectroscopy (AAS) is an analytical testing method used to determine the concentration of specific metal elements in a sample such as lead, cadmium, and iron.

This analysis will be done to check the concentration of heavy metals in the sample.

The test cells samples were prepared as follows. The flasks of 500 mL which provided one bottle weighing between 10g to 150g were used. About 1000 mL of known acid was weighed. A known weight (20g) of the steel sample balls were placed in 1000 mL and diluted with 2 $\times$ known concentration HNO₃ and prepared for AAS analysis on a Perkin Elmer Atomic Fluorescence Spectrophotometer model 50-200 were tested on.



Photo 1: Burning process of tyre waste used



Photo 2: (eye closed state)



Photo 2: grassland of grassland becomes eroded area



Fig 4. *Artemia* (larval stage)



Figure 2: Presence of promising barrow-like crop fields.



Flan 6. [Fertilizantes.com](http://www.fertilizantes.com)

3.3.3 Statistical Analysis

Data obtained were subjected to analysis of variance (ANOVA) using SPSS and means were separated using Duncan multiple Range Test with statistical significance of 5% [17].

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1. POLY AROMATIC HYDROCARBON CONSTITUENTS PRESENT IN THE SKIN OF TYPE, KENGENE AND PEROL-SENGED COW MEAT

The Poly Aromatic Hydrocarbon constituents present in the skin of type, kengen and perol-senged meat are presented in Table 4.1.

Compound	Type Sking	Kenene Sking	Perol Sking
Naphthalene	2.14×10^{-07}	1.34×10^{-07}	1.41×10^{-07}
Acenaphthylene	1.89×10^{-07}	1.11×10^{-07}	1.89×10^{-07}
Acenaphthene	ND	9.14×10^{-07}	8.79×10^{-07}
Phenanthrene	2.10×10^{-07}	1.41×10^{-07}	2.39×10^{-07}
Benzo(a)Pyrene(baP)	9.14×10^{-07}	9.36×10^{-07}	9.34×10^{-07}

Table 4.1 Poly Aromatic Hydrocarbon in the skin of type, kengen and perol-senged.

Therapeutic dose of 100 mg	1.136 ± 0.007	0.114 ± 0.007	2.250 ± 0.007
Therapeutic dose of 200 mg	1.079 ± 0.007	0.115 ± 0.007	2.193 ± 0.007
Therapeutic dose of 300 mg	1.096 ± 0.007	0.114 ± 0.007	2.210 ± 0.007

There are no significant differences between the tested doses. Also, in the tested values and different regimens are equivalent, different ($p < 0.05$).

4.1.15 Ergothioneine

Ergothioneine is an ergane compound with formula $C_{10}H_{16}N_2O_4$. It is the simplest polycyclic aromatic heterocycle and is a water-soluble solid with a characteristic odor that is described as reminiscent of wet soil (9) (10) (11).

The study for all the samples type, however, and period ranged from 1.00000 mg/kg to 24.00000 mg/kg with period ranges being the highest value. There were significant differences in the values obtained (0.05) and the way not be associated with different regimens tested that were used.

State exposure of human to ergothioneine by inhalation exposure and based around is associated with burning, storage, storage in the food and, in addition, nutritional therapy. According to the European Food Safety Authority (EFSA, 2014), the permitted value for ergothioneine is 0.000 mg/kg. The value obtained is below the maximum concentration not to be consumed due to its accumulation in the body.

4.1.3 Acetylcholine

Acetylcholine is a polysaccharide around hydrocarbon composed of three kinds of amino acids: two saturated amino acids and one five saturated amino acids. Although acetylcholine is not classified as a protein, human consumption of it is very important to maintain human health with other PMA and may contribute to respiratory effects, headache and fatigue.

The result for all the samples (i.e. human and plant) ranged from 1.01 to 1.02 mg/kg with plant having the highest value. There were significant differences in the values obtained ($p < 0.05$) and may not be measured with different sampling methods that reported.

European Food Safety Authority (EFSA, 2003) recommended maximum permitted level of 0.2 mg/kg. The value obtained is lower but continuous consumption will cause the accumulation in the body which can pose threat to human health.

4.1.3 Acetylcholine

Acetylcholine is a white crystalline solid derived primarily from choline and acetic anhydride and is composed of two parts. Acetylcholine may be produced or obtained or extracted through HPLC. The major way to produce it is by extraction. Acetylcholine may cause poisoning if consumed in large quantity. It can cause difficulty breathing.

The result for all the samples (i.e. human and plant) ranged from 0.14 to 0.15 mg/kg with plant having the highest value and it is not detected in fish sample. There were significant differences in the values obtained ($p < 0.05$).

EFSA advised human consumption recommended maximum level of 0.2 mg/kg (FAO, 2003). Although the value obtained is lower, frequent consumption is not advised.

4.1.4 Phenol

Reaction is a polymeric material. Hydrocarbon that is insoluble in a given solvent is organic solvent. It forms that some long chain of being exposed to.

The results obtained for Reaction showed a range of 0.0000002 to 0.0000002 with sample 17 (green) having the highest value and sample 16 (grey) having the lowest. There were significant differences in the values obtained ($p < 0.05$) and they may not be associated with the different sampling methods that were used.

4441 (World Health Organization) recommended particulate limit of 0.0000002 (WHO 2000). The results showed a lower but continuous consumption should be avoided due to the of chronic toxicity.

4.1.2. Benzene

Benzene is a polymeric material. Hydrocarbon and the level of benzene consumption of organic material consumption between 0.0000002 (0.0000002) and 0.0000002 (0.0000002). The consumption consumption can be found in and for release levels and more likely, especially, given more. In that, hydrocarbon, were commonly known as 0.0000002, more with and less as 0.0000002, leading to maximum and eventually more. It is found as a range of 0.0000002 to 0.0000002, in 0.0000002 is significant to reduce 0.0000002 due to its exposure with the organic hydrocarbon consumption (0.0000002). Apart from its chemical exposure in development and application, the chemical also represents the chemical system.

The results obtained for benzene showed a range of 0.0000002 to 0.0000002 with sample 17 (green) having the highest and sample 16 (grey) having the lowest. There are significant differences in the values obtained ($p < 0.05$). It is significant to reduce consumption due to its exposure with the organic hydrocarbon consumption (0.0000002). Apart from its chemical exposure in development and application, the chemical also represents the chemical system.

The benzene (benzene) has a maximum limit of 1mg/kg (wt: 200%) of monomer in some products. Therefore, the result obtained is lower but it is also a risk that there is continuous consumption as a result of this adulteration.

4.14: Isobutyl (I.B.) (mg/glycose)

Isobutyl (I.B.) (mg/glycose) is a polyaromatic aromatic hydrocarbon (PAH), one of 16 PAHs generally mentioned in studies of environmental exposure and air pollution. The compound has found in some foods, monomers and/or oligomers. The substance can be absorbed into the body by inhalation of its aerosol and through the skin. This substance is possibly carcinogenic to humans.

The result obtained isobutyl (I.B.) (mg/glycose) showed a range of 0.24 (0.00) to 0.42 (0.02) with sample 1 (purple) being the highest and sample 8 (pink) being the lowest. There are significant differences in the values (Kruskal-Wallis 0.01) and this may not be associated with the different sampling locations but more with the compound's nature as there were monomers and oligomers. The substance can be absorbed into the body by inhalation of its aerosol and through the skin.

Multi-benzene (benzene) (PAH) has a maximum limit of 1.2mg/kg (WFF:200%) of B in some products. Therefore, the result obtained is close to this range and continuous consumption of this can lead to health issues since they are carcinogenic.

4.15: Benz (g.L) (perylene (B) (g.L) (B)

Benz (g.L) (perylene) is a polyaromatic aromatic hydrocarbon. It is a solid white crystal like with. One of the most common types (benz (g.L) (perylene) are aromatic hydrocarbons through burning, concentrated in soot. It is carcinogenic (classified as a known, probable and

2,3,5, 6-phenyl-1,4-bisulfone has been classified as *probably carcinogenic to humans*, group 2B (IARC group 2B).

No epidemiological studies on human exposure to 2,3,5,6-tetrafluorobenzene have been conducted. First, given limited TQPs, there was a component of complex chemical mixture and second, it is not in the environment.

4.1. POLY AROMATIC HYDROCARBON CONSTITUENTS PRESENT IN THE FLAT OF TEEB, KIRKOWITZ AND PETERL SIGNED-COW MEAT

Only traces of hydrocarbon constituents present in the flat of tee, kirkowit and peterl, complexed compounds in Table 4.2.

Table 4.3 Poly aromatic Hydrocarbon constituents present in the flat of tee, kirkowit and peterl signed meat.

Compound	Teeb (ug/g)	Kirkowit (ug/g)	Peterl (ug/g)
Naphthalene	0.11 ± 0.02 ^a	NA	NA
Acenaphthylene	0.00 ± 0.00 ^a	0.02 ± 0.00 ^a	0.00 ± 0.00 ^a
Acenaphthene	0.00 ± 0.00 ^a	NA	NA
Fluorene	NA	0.00 ± 0.00 ^a	0.00 ± 0.00 ^a
Indene	NA	0.02 ± 0.00 ^a	0.01 ± 0.00 ^a
Benz[a]Pyrene	NA	0.11 ± 0.02 ^a	0.14 ± 0.02 ^a

of Pyrus

Values are means $\pm$ standard deviation and standard deviation. Values in the same column with different superscripts are significantly different ($p < 0.05$).

4.2.2. Nephelium

Nephelium is an organic compound with formula $C_{10}H_{10}$. It is the simplest polycyclic aromatic hydrocarbon and is a white crystalline solid with a characteristic odor that is described as reminiscent of kerosene or kerosene-type fuel.

The mean observed for all samples ranged from 5.0 to 6.17 $\pm$ 0.02 with sample 6 (Pyrus) having the highest and sample 5 (quercus) and sample 10 (Pinus) the lowest. There are significant differences in the values observed ($p < 0.05$). The results show that the quantity of the compound decreases compared to that found in the list of the highest and lowest.

Acute exposure of humans to nephelium by inhalation, ingestion, and dermal contact is associated with respiratory irritation, damage to the liver, and to other, non-specific damage to the central nervous system.

According to the European Food Safety Authority (EFSA, 2004), the permitted value for nephelium is 1.0 mg/kg. The mean observed is lower than maximum consumption can lead to affect due to low accumulation in the body.

4.2.2 Acetylcholine

Acetylcholine is a polysaccharide, viscous liquidation composed of three bound molecules one ring and one free terminal ring. Though acetylcholine is not classified as a protein based substance on its own, it shares structural features with other PABs and may contribute to respiratory distress, bronchitis and lung cancer.

The results obtained for acetylcholine showed a range of 10144.892 to 34568.552 with sample A (type1) being the highest and sample C (type3) being the lowest. There are significant differences in the values obtained ($p < 0.05$). This result shows that the quantity of the substance decreases compared to that found in the sites of the original three samples.

Though acetylcholine is not classified as a protein based substance on its own, it shares structural features with other PABs and may contribute to respiratory distress, bronchitis and lung cancer.

European Food Safety Authority (EFSA, 2006) recommended maximum permissible level of 67 mg/kg. The value obtained is lower but continuous consumption can cause low concentrations in the body which can pose threat to human health.

4.2.3 Acetylcholine

The results obtained for acetylcholine showed a range of 161 to 34568.552 with sample A (type1) being the highest value and sample H and C are lowest. There are significant differences in the values obtained ($p < 0.05$). Acetylcholine may be poisonous if inhaled or absorbed through skin. The major way of low decrease in utilization. Acetylcholine may cause vomiting if swallowed or large quantity is absorbed either.

FAO/WHO (2006) recommended maximum permissible level of 67 mg/kg (FAO 2001). Although the value obtained is lower, frequent consumption is not advised.

4.1.4 Phenothrene

Phenothrene is a polycyclic aromatic hydrocarbon (PAH) with formula $C_{14}H_{10}$, consisting of three fused benzene rings. It is a solid, orange-brown solid, but can also appear yellow. Phenothrene can affect the body system when inhaled. It is known to irritate the skin and eyes. If it is combined with Hexamethine it is expected to undergo a sort of chemical reaction, consistent with theory.

The result obtained for phenothrene ranges from 90 to 1000000 with sample 1 ignored during the trial. The result shows that the quantity of the constituent elements compares to that found in the list of the target test sample.

4.1.5 Anthracene (a.k.a)

The result obtained for phenanthrene ranges from 90 to 1000000 with sample 1 ignored during the trial. The result shows that the quantity of the constituent elements compares to that found in the list of the target test sample.

Hexamethine is a chemical compound consisting of six carbon atoms, but of different configurations resulting in the same molecular formula. It is a solid, colorless to yellow and very flammable. The 1,3,5-**CD4** Integral Unit Information File Information System (IUIS) reports several information items (PII) and is being updated by the professional IUIS (PII) (1990-1992).

4.1.6 BenzilPyrene

The result obtained for phenanthrene ranges from 90 to 1000000 with sample 1 ignored during the trial. The result shows that the quantity of the constituent elements compares to that found in the list of the target test sample.

The Integral Unit Information File (IUIS) of Integral Unit Information System (IUIS) reports several information items (PII) and is being updated by the professional IUIS (PII) (1990-1992). The result obtained is from the 1,3,5-**CD4** Integral Unit Information System (IUIS) (1990-1992) and shows that the constituent elements compare to that found in the list of the target test sample.

4.3. POLY AROMATIC HYDROCARBON CONSTITUENTS PRESENT IN THE COATING OF COW HIDE (POTHAQ)

Only aromatic hydrocarbon constituents present in the coating of cow hide (Pothaq) are presented in Table 4.3.

Table 4.3 Only aromatic hydrocarbon constituents present in the coating of cow hide (Pothaq)

Compound	Hex	1Hex	4Hex	7Hex
Naphthalene	0.0440.0004	0.0440.0004	0.0440.0004	0.0440.0004
Acenaphthylene	0.0710.0007	0.0710.0007	0.0710.0007	0.0710.0007
Acenaphthene	0.0710.0007	0.0710.0007	0.0710.0007	0.0710.0007
Fluorene	0.0110.0001	0.0110.0001	NA	NA
Benzo(b)fluorene	0.0110.0001	0.0110.0001	NA	NA
Benzo(a)Pyrene	0.0110.0001	0.0110.0001	NA	NA
Indeno(1,2,3-cd)Pyrene	0.0110.0001	0.0110.0001	NA	NA
Dibenz(a,h)Anthracene	0.0110.0001	0.0110.0001	NA	NA

There are many *not* significant differences in empirical findings. (But in the same volume with different signatories and especially different χ^2 (2.2)).

4.1.1. *Sample Size*

The result obtained for all samples ranged from 1.07 to 1.09 to 1.11 with the sample C (10 hours) having the highest value and sample B (7.5 hours) having the lowest value. There are significant differences in the values obtained ($p < 0.05$) and they may not be associated with the different working period. These results show that adaptation continuously depends on the human working period.

According to the European Food Safety Authority (EFSA, 2004), the permissible value for adaptation is 1.05 mg/kg. The value obtained is lower than maximum recommended and that means due to his accumulation in the body.

4.1.2. *Sample Flow*

The result obtained for sample flow ranged a range of 0.24 to 0.27 to 0.30 with sample C (10 hours) having the lowest value and sample A (5 hours) having the highest value. There are significant differences ($p < 0.05$) in the values obtained and they may not be associated with the different working period.

European Food Safety Authority (EFSA, 2005) recommended maximum permissible limit of 0.2 mg/kg. The result obtained is higher which is it not to estimate as working should be done effectively to reduce the accumulation of the substance.

4.1.3. *Sample Time*

The result obtained for sample time ranged a range of 0.04 to 0.07 to 0.11 with sample B (7.5 hours) having the lowest value and sample A (5 hours) having the highest value. There are significant differences in the values obtained ($p < 0.05$) and they may not be associated with the different working period.

4.4. POLY CHLORINATED BIPHENYL CONSTITUENTS REPORT IN THE MEAT
OF THE VARIOUS RANGES COW MEAT

Table 4.4: Poly chlorinated Biphenyl constituents present in the diet of goat, lambs and
pork raised meat.

Compound	Goat Diet	Lambs Diet	Pork Diet
1,1'-Biphenyl(1,2,3,4,4',5'- hexachloro-PCB 118)	0.005±0.002	0.005±0.002	0.01±0.002
1,1'-Biphenyl(1,2,3,4,5,5'- pentachloro-PCB 110)	0.004±0.002	0.006±0.002	0.11±0.002
1,1'-Biphenyl(1,2,3,3',4,4',5'- hexachloro-PCB 116)	0.005±0.002	0.004±0.002	0.005±0.002
1,1'-Biphenyl(1,2,3,4,4',5,5'- hexachloro-PCB 117)	0.012±0.002	0.003±0.002	0.018±0.002
1,1'-Biphenyl(1,2,3,4,4',5,5'- heptachloro-PCB 120)	0.027±0.002	0.023±0.002	0.023±0.002
1,1'-Biphenyl(1,2,3,4,4',5,5',6'- heptachloro-PCB 121)	0.0007±0.002	0.009±0.002	0.009±0.002
1,1'-Biphenyl(1,2,3,4,4',5,5',6'- heptachloro-PCB 124)	0.003±0.002	0.003±0.002	0.003±0.002
1,1'- Biphenyl(1,2,3,3',4,4',5,5',6'-	0.004±0.002	0.007±0.002	0.007±0.002

acetaldehyde-PCB (PCB 104)

1,1'-Biphenyl(1,2,3,4,1',2',3',4')- 0.053±0.002^a 0.051±0.004^a 0.050±0.002^a

heptachlor- (PCB 187)

1,1'-Biphenyl(1,2,3,5,4',3',5',4')- 0.005±0.002^a 0.004±0.002^a 0.004±0.002^a

heptachlor- (PCB 170)

Values are means of triplicate determinations ± standard deviation. Blank in the parentheses will differ by approximately 10% (standard deviation) ($p < 0.05$).

4,4'-1,1'-Biphenyl(1,2,3,4,5'-Acrodien-3-PCB18)

Endosulfan Sulfate or PCE 118, is a chemical compound belonging to the group of poly(chlorinated biphenyls) PCBs. This is a synthetic organic substance composed of two linked benzene rings (biphenyl) with varying numbers of chlorine atoms. In PCE 118, ten chlorine atoms are attached at specific positions on the biphenyl structure: 2,2',3,4,4', and 5,5'.

In terms of health effects, PCE 118 is associated but has been associated with a range of adverse outcomes. These include adverse respiratory, neurodevelopmental effects, immune system suppression, and hormonal disruption. It is considered a priority chemical candidate for the International Agency for Research on Cancer (IARC).

Due to its wide geographic and environmental presence, PCE 118 is found in widely reported in many countries. It is listed under the Stockholm Convention on Persistent Organic Pollutants, and its presence in the environment, food, and manufactured products is subject to international monitoring and control.

The most studied for Li^+ -Epigallocatechin-3-gallate (EGCG)-induced a range of mitochondrial dysfunction with sample y_1 subjects having the highest values and sample y_2 subjects having the lowest values. There are no significant differences in the values obtained for the variables may not be associated with the different sampling methods that were used. Under that, but not for individual PCE 118 or PCE 118 has positive guidance on food intake.

4.4.2. Li^+ -Epigallocatechin-3-gallate (EGCG)-induced (PCE 118)

Endosulfan Sulfate or PCE 118, is a specific compound within the group of poly(chlorinated biphenyls) PCBs, is the case of PCE 118, five chlorine atoms are attached at the 2,2',3,4,4', and 5,5' positions of the biphenyl molecule. Like other PCBs, PCE 118 is chemically stable, resistant to biodegradation, and bioaccumulative.

response to WBC. The other WBC components, too, have induced a range of pleiotropic health effects. These include potential impact on the liver, immune system, reproductive system, and endocrine system. There is also evidence that WBC may disrupt neurological development and contribute to cancer risk.

The results obtained for $3,3'$ -Bisphenol[2,2'-A,2,2'-bis(4-oxazolin)](10,000) showed a range of 0.14(0.02) to 0.23(0.02) with sample 1' (green) having the highest value and sample 4p(4q) having the lowest. There are no significant differences in the values obtained (0.05) and this could be associated with the different polymer molecules that make up it.

The authors are grateful to the referees for their constructive comments.

[illegible]

1,1-dimethyl-2-ethyl-3-oxo-2-butene (PBTDA), is a member of the polyfunctional hydrocarbons (PDA), a group of synthetic organic materials now widely used in industrial applications. PBTDA consists of two functional vinyls (a hydroxyl-terminated polymeric chain) connected at the 2, 3, 3', 4, 4' and 5' positions.

Exposure to PCB 118 has been associated with a variety of health effects, including disruption of the endocrine system, liver damage, immune system suppression, reproductive and developmental toxicity, and possible carcinogenicity. Because of these risks, PCB 118 and other dioxin-like PCBs are closely monitored in food supplies, especially in fish, meat, and dairy products.

The analysis of the *IL-6* gene (*IL6*, accession #U03334) showed a range of polymorphisms in the 5'UTR with sample 1 (pilot) having the highest value and sample 6 (post 42 hours) having the lowest. There were no significant differences in the range between pilot and post 42 hours and the analysis of the *IL-6* gene showed no significant differences between pilot and post 42 hours.

The t -test for $H_0: \beta = 0$ for some polynomial p in $\text{Eqn}(1)$ (Equation 2.10) is 2060.2111. The value obtained is far from the test that zero should be ruling it out for consideration.

4.4.4.1: $\text{Eqn}(12.7.4.1)$ (Equation 2.11)

Equation (2.11) is a well-known component of the polynomial system (2.11) (Eqn). (2.11) has an affine space defined at 1, 2, 3, 4, 5, and 6 for the system structure. (2.11) is associated with some algebraic field effects. It has been shown to affect the system's growth function, and because equation (2.11) has no constant effect. While it is not clear the impact is even after (2.11) component, its presence and absence cannot significantly affect (2.11) (Eqn) system.

The test obtained for $\text{Eqn}(12.7.4.1)$ (Equation 2.11) showed a value of 2060.2111 as 1.22-0.001 with sample (2) (sample) being the highest sample and sample (1) being the lowest. There are significant differences in the value. (2.11) and this may not be associated with the different sample method. The value is 2060.2111.

The t -test obtained for the test that $\beta = 0$ is 2060.2111 (Equation 2.10). The value obtained is far from the test.

4.4.5.1: $\text{Eqn}(12.7.4.2)$ (Equation 2.12)

(2.12) (Eqn), along with other (2.12), were tested to identify constant (2.12) (Eqn). (2.12) (Eqn) shows a form of polynomial system (2.12) (Eqn) is associated with some algebraic field effects. It has been shown to affect the system's growth function, and because equation (2.12) has no constant effect. While it is not clear the impact is even after (2.12) component, its presence and absence cannot significantly affect (2.12) (Eqn) system. (2.12) (Eqn) is associated with the different sample method. The value is 2060.2111.

The most elevated for *L. P. Bignoni*(12.7,544,1.9° elevation,PT 1419) showed a range of 0.02160002 to 0.02160002 with sample 4 being the highest value and sample 10 being the lowest. There are significant differences in the values obtained (p<0.05). The 1419 summer tree tree stand and tree products is 100g/g (transport 1 summer 2006). The values obtained are not close to this.

446. *L. Pignoni*(12.7,544,2° Elevation)(PCB 104)

Commonly known as PCB 104, is a synthetic compound. It consists of one benzene ring connected by a single bond to a biphenyl structure, with seven chlorine atoms attached at specific positions. 2,2',3,3',4,4',5'- and 6- This high degree of chlorination makes PCB 104 one of the most heavily chlorinated congeners in the PCB group. Like other highly chlorinated PCBs, PCB 104 is extremely stable and resistant to degradation in the environment. It is also highly lipophilic, meaning it tends to accumulate in fatty tissues of living organisms, including fish, marine mammals, birds, and humans. Due to its bio-accumulative and bio-magnification properties, PCB 104 is associated with adverse environmental effects.

Although PCB 104 is considered a non-dioxin-like PCB, it can still pose health risks through chronic exposure. While it has not caused the same toxic problems as dioxin-like PCBs, it has been associated with potential effects on liver function, the endocrine system, and developmental processes. The study presented for *L. Pignoni*(12.7,544,2.9° Elevation,PT 2019) showed a range of 0.02160002 to 0.02160002 with sample 8 being the highest, and sample 8 and 1° Elevation and 1000g/g being the lowest. There are no significant differences in the values obtained (p>0.05). There's no specific limit for PCB 104 but most consumption should be avoided.

447. *L. Pignoni*(12.7,544,2.9° Elevation)(PCB (PCB 104))

Commonly known as P.R. 194, it is one of the highly phloemated members of the polydormant type (P.R.s). It consists of a lignary structure (a pair of linked internode rings with eight dormant axons attached at the 2,2',3,3',4,4', and 6 positions. The high degree of dormancy makes P.R.194 extremely water and nutrient sensitive in the stemwood.

Although P.R. 194 is a non-dormant P.R., it still grows highly sensitive due to its long-term presence in the environment just provided to succinate, to forest and forest to forest. Chronic exposure has been associated with effects on forest health, but no other system, and possibly developmental and ecological responses. The most common for *L.L.*, **Spine(1,2,3,3',4',5')** (dormant) (P.R.194) consists of a single of linked internode rings with eight dormant axons having the highest (a single ring) and 6' (dormant and paired) being the lowest. Thus, no significant difference in the total amount of indicated response time is set for P.R. 194.

4.4.3.1. **Spine(1,2,3,4,5,5',6')spines (P.R.197)**

Commonly known as P.R. 197, it is composed of two linked internode rings with various number and positions of dormant axons. P.R. 197 consists of two dormant axons located at the 2,2',3,3',4,4', and 6 positions, making it one of the more heavily phloemated P.R.s. Commonly, P.R. 197 is lignified, meaning it tends to succinate in the long hours of being exposed. The above ring succinate and the ring(s) through the first three, resulting in high concentrations of top positions, including leaves.

Although P.R. 197 is considered a non-dormant P.R., it still grows (grows) a little. Exposure to P.R. 197 has been associated with forest canopy reduction, reduction in growth, and effects on the immune and reproductive systems. It is likely to persist in the environment and succinate in the long-term if a source for long-term exposure exists.

The result obtained for **LT-Spinner(11'11'11'44'55')** condition generally showed a range of 0.021 to 0.02 to 0.026 to 0.02 with sample 4 being being the highest and sample 1, generally being the lowest. There are no significant differences in the values obtained by 0.021 and they may not be interconnected with the different bagging material that was used. The individual treatment that is as for 0.02 to 0.02.

4.4.3 LT-Spinner(11'11'11'44'55'-acetabular (PCE 10)

Experiments related to as 0.02 to 0.02, it contains all a higher level material, two layers were covered by a single coat with eight layers were conducted at the 11'11'11'44'55' and 5' positions. This is an eight layers system, 0.02 to 0.02 is one of the most important and the most important program in the program. With 11'11'11'11' is a wide range of bagging. It may still give health concerns. The result obtained for **LT-Spinner(11'11'11'44'55')** shows 0.021 to 0.021 (from sample 4 to sample 1) (given). No individual treatment that is as for 0.02 to 0.02.

4.4.3 LT-Spinner(11'11'11'44'55'-bagging (PCE 10)

Experiments related to as 0.02 to 0.02, it contains all a higher level material, two layers were covered by a single coat with eight layers were conducted at the 11'11'11'44'55' and 5' positions. This is an eight layers system, 0.02 to 0.02 is one of the most important and the most important program in the program. With 11'11'11'11' is a wide range of bagging. It may still give health concerns. The result obtained for **LT-Spinner(11'11'11'44'55')** shows 0.021 to 0.021 (from sample 4 to sample 1) (given). No individual treatment that is as for 0.02 to 0.02.

The result obtained for **LT-Spinner(11'11'11'44'55'-bagging (PCE 10)** shows a range of 0.021 to 0.02 to 0.026 to 0.02 with sample 4 being being the highest and sample 1, generally being the lowest. There are no significant differences in the

values obtained previously and they may not be associated with the different sampling methods that have used. The statistical treatment described in the 1992 [15].

4.1. POLY CHLORINATED BIPHENYL CONSTITUENTS PRESENT IN THE FLESH OF THE CARBOUS EDGED COYALAT

Table 4.5: The Poly (Chlorinated Biphenyl) constituents present in the flesh of the coy-
alatus and parent coyolatus

Compound	Two Day	Seven Day	Two Day
1,2'-Biphenyl(1,1,1,4,4,4)-hexachloro- (PCH 10)	0.000000000	0.000000000	0.000000000
2,2'-Biphenyl(1,1,1,4,4,4)-hexachloro- (PCH 10)	0.000000000	0.000000000	0.000000000
1,2'-Biphenyl(1,1,1,4,4,4)-hexachloro- (PCH 10)	0.000000000	0.000000000	0.000000000
1,2'-Biphenyl(1,1,1,4,4,4)-hexachloro- (PCH 10)	0.000000000	0.000000000	0.000000000
1,2'-Biphenyl(1,1,1,4,4,4)-hexachloro- (PCH 10)	0.000000000	0.000000000	0.000000000
1,2'-Biphenyl(1,1,1,4,4,4)-hexachloro- (PCH 10)	0.000000000	0.000000000	0.000000000
1,2'-Biphenyl(1,1,1,4,4,4)-hexachloro- (PCH 10)	0.000000000	0.000000000	0.000000000
1,2'-Biphenyl(1,1,1,4,4,4)-hexachloro- (PCH 10)	0.000000000	0.000000000	0.000000000
1,2'-Biphenyl(1,1,1,4,4,4)-hexachloro- (PCH 10)	0.000000000	0.000000000	0.000000000
1,2'-Biphenyl(1,1,1,4,4,4)-hexachloro- (PCH 10)	0.000000000	0.000000000	0.000000000
1,2'-Biphenyl(1,1,1,4,4,4)-hexachloro- (PCH 10)	0.000000000	0.000000000	0.000000000

1.7. Epsilon(1.7.1.7.4.4) 1.7.

Accepted:

Accepted:

Accepted:

Accepted: (1.7.1.7.4.4)

1.7. Epsilon(1.7.1.7.4.4) 1.7.

Accepted:

Accepted:

Accepted:

Accepted:

Tables are based on *Empirical data* (see *Table 1* in the text) and *Table 2* in the text. The data are based on the results of the *Empirical data* (see *Table 1* in the text) and *Table 2* in the text. The data are based on the results of the *Empirical data* (see *Table 1* in the text) and *Table 2* in the text.

4.1.1.1' **Significance(1,73,4,F) Assesses (PCB III)**

The result showed for 1,1'-Dibiphenyl(1,73,4,F) Assesses (PCB III) showed a range of statistically significant with sample 6 (year) being the highest and sample 8 (summit) being the lowest. There are significant differences in the values obtained.

PCB III is toxic and has been associated with a range of adverse outcomes. These include endocrine disruption, immunosuppression effects, immune system suppression, and potential carcinogenicity. It is considered a priority human concern by the international agency for Research on Cancer (IARC).

The III treatment had the most and least products in 2004 (European Commission 2004). The values obtained are as shown in the.

4.1.1.1' **Significance(1,2,4,5,5'-parameters-PCB III)**

The result showed for 1,1'-Dibiphenyl(1,2,4,5,5'-parameters-PCB III) showed a range of 6 (summit) to 4,264,602 with sample 6 (year) being the highest and sample 8 (summit) being the lowest. There are significant differences in the values obtained.

regional PCB III. The PCB III program, has been linked to a range of adverse health effects. These include potential impacts on the liver, immune system, reproductive system, and endocrine system.

The III treatment had the most and least products in 2004 (European Commission 2004-2011). The values obtained are as shown in the.

4.1.1.1' **Significance(1,2,3,7,4,F) Assesses (PCB 150)**

The result obtained for **LT-Expans(1000/4)** (see table 49C) has showed a range of test values as 4.000000E-01 with sample 1 (control) being the highest and sample 4 (PCV) being the lowest. There are no significant differences in the values obtained.

Expans in F.R. 110 has been associated with a variety of health effects, including disruption of the endocrine system, liver damage, immune system suppression, reproductive and developmental toxicity, and potential carcinogenicity. Because of these risks, F.R. 110 and other fluids like PCBs are closely monitored in food supplies, especially in fish, meat, and dairy products.

The 10th assessment letter for trans and trans products is being a (see page 1) (see table 49B). The values obtained are as shown in this.

4.24 LT-Expans(1000/4/10) (see table 49C)

The result obtained for **LT-Expans(1000/4/10)** (see table 49C) has showed a range of test values as 4.000000E-01 with sample 1 (control) being the highest and sample 4 (PCV) being the lowest. There are significant differences in the values obtained and they may not be associated with the different dosing conditions that were used. PCBs are associated with various adverse health effects. It has been shown to affect the immune system, liver function, and hormone regulation, and it may have neurotoxic effects. While it is less Acute-like compared to some other PCB congeners, its persistence and bioaccumulation capabilities significantly increase PCB exposure concerns. The 10th assessment letter for trans and trans products is being a (see page 1) (see table 49B). The values obtained are as shown in this.

4.25 LT-Expans(1000/4/10) (see table 49C)

The result obtained for **LT-Expans(1000/4/10)** (see table 49C) has showed a range of test values as 4.000000E-01 with sample 1 (control) being the highest and sample 4 (PCV) being the lowest. There are significant differences in the values obtained (see table 49B).

The 111 treatment level for both soil types produced a 100% (100/100) germination (Table 1).
The values obtained are not listed in this table.

4.14 11' *Epilene*(12' 3.14 7' 8.4)epilene(PCB 134)

Compound tested as PCB 134, is a synthetic compound, it consists of two aromatic rings connected by a single bond to biphenyl structure, with seven chlorine atoms substituted at specific positions (1,2,3,4,6,7, and 8).

The result obtained for 11' *Epilene*(12' 3.14 7' 8.4)epilene-epilene(PCB 134) showed a range of 0.000-0.001 to 0.000-0.001 with sample A (type) having the highest and sample B (herbicide) having the lowest. There are no significant differences in the values obtained ($p > 0.05$). There's no specific limit for PCB 134 for crop consumption because it is not listed.

4.15 11' *Epilene*(12' 3.14 7' 8.4)epilene(PCB 136)

The result obtained for 11' *Epilene*(12' 3.14 7' 8.4)epilene-epilene(PCB 136) showed a range of 0.000-0.001 to 0.000-0.001 with sample A (type) having the highest value and sample C (herbicide and pesticide) having the lowest. There are significant differences in the values obtained ($p < 0.05$). No withdrawal maximum limit is set for PCB 136.

4.16 11' *Epilene*(12' 3.14 7' 8.4)epilene(PCB 137)

The result obtained for 11' *Epilene*(12' 3.14 7' 8.4)epilene-epilene(PCB 137) showed a range of 0.000-0.001 to 0.000-0.001 with sample A (type) having the highest and sample C (pesticide) having the lowest. There are no significant differences in the values obtained ($p > 0.05$) and they may not be associated with the different spraying materials that were used. No withdrawal maximum limit is set for PCB 137.

4.18 11' *Epilene*(12' 3.14 7' 8.4)epilene(PCB 140)

The result obtained for **U2_Eggsac(11'15'44.58)** indicates a value smaller than sample A to sample I (p=0.6). There is no significant difference in the values obtained given 95% the individual treatment level is set for 0.05 (95%).

4.2.8.11-Spleeny(11'15'44.58)positive- (PCR 17)

The result obtained for **U2_Eggsac(11'15'44.58)** indicates a value smaller than sample A to sample I (p=0.6). There is no significant difference in the values obtained given 95% the individual treatment level is set for 0.05 (95%).

Table 11.11.4.17:

residuals (P < 0.0001)

1.0 0.0000000 0.0000000 0.0000000 0.0000000

Table 11.11.4.18:

residuals (P < 0.0001)

1.0 0.0000000 0.0000000 0.0000000 0.0000000

Table 11.11.4.19:

residuals (P < 0.0001)

1.0 0.0000000 0.0000000 0.0000000 0.0000000

Table 11.11.4.2:

residuals (P < 0.0001)

residuals (P < 0.0000000) 0.0000000 0.0000000 0.0000000 0.0000000

0.00

Values are means of individual determinations & standard deviation. Blank in the above columns imply different superscripts are significant at (overall) $p < 0.05$.

443: 11' *Agave*(U7344)' description (PCB III)

The seed obtained for 11' *Agave*(U7344)' description (PCB III) obtained a range of germination at 25°C/20°C, with sample A (0 hour) being the highest and sample (15/12 hour) being the lowest. There are significant differences in the values obtained (p < 0.05). This shows that *P* is constantly increases with increase in sowing period.

PCB III is toxic and has been associated with a range of adverse reactions. These include systemic disruption, acute developmental effects, immune system suppression, and potential carcinogenicity. It is considered a priority human carcinogen by the International Agency for Research on Cancer (IARC).

The 40% maximum limit for new and used products is being proposed (maximum 20%). The values obtained are not close to this.

443: 11' *Agave*(U745)' germination (PCB III)

The seed obtained for 11' *Agave*(U745)' description (PCB III) showed a range of germination at 25°C/20°C, with sample B (24 hours) being the highest and sample A (0 hour) being the lowest. There are no significant differences in the values obtained (p > 0.05).

Agave is a PCB III. The PCB III compound, has been linked to a range of adverse health effects. These include potential impacts on the liver, immune system, reproductive system, and endocrine system.

The 10% maximum limit for new and used products is being. The values obtained are not close to this.

443.11-Spline(11.33.4.4) -Description-(PCB158)

Spline 11-PCB 158 has been associated with a variety of health effects, including disruption of the endocrine system, liver damage, immune system suppression, reproductive and developmental toxicity, and potential carcinogenicity. Studies of liver 158, PCB 158 and other dioxin-like PCBs are closely monitored in food systems, especially in fish, meat, and dairy products.

The study material for 11-Spline(11.33.4.4) -Description-(PCB158) material is made up of 60000000 or 60000000 with sample 0 (0) being the highest and sample 4 (4) being the lowest. There are no significant differences in the values themselves (158).

The (11) material has been used and more problems in 443.11-Spline(11.33.4.4) -Description-(PCB158). The values obtained are not close to the

444.11-Spline(11.34.1.1) -Description-(PCB158)

PCB 158 is associated with various adverse health effects, if the dose is high or affect the immune system, liver damage, and immune suppression, and it may have carcinogenic effects. While it is less than that compared to some other PCB congeners, its persistence and abundance can have significant impacts on food PCB levels in humans.

The study material for 11-Spline(11.34.1.1) -Description-(PCB158) material is made up of 60000000 or 60000000 with sample 0 (0) being the highest and sample 6 (6) and 7 (7) being the lowest. There are significant differences in the values themselves (158) and they may not be associated with the different congeners material that were used.

The (11) material has been used and more problems in 444.11-Spline(11.34.1.1) -Description-(PCB158).

The values obtained are not close to the

443: 11'Spline(12,14,5)Asymptotic(PCR10)

Continuity tested at PCR 104, is a continuity segment. It consists of two linear segments connected by a single bend by hyperbol segments with some other segments connected at specific positions. 12,14,5, and 5.

The result obtained for 11'Spline(12,14,5)Asymptotic(PCR10) showed a range of 0.0114001 to 0.0114002 with sample 4. It is not being the highest and sample 1012 being the lowest. There are significant differences in the values obtained 0.011. This shows that PCR segment decreases with increase in working period. There's no specific limit for PCR 104 (0.0114001) that is a presumption should be avoided.

444: 11'Spline(12,17,4,2,2)Asymptotic(PCR10)

The result obtained for 11'Spline(12,17,4,2,2)Asymptotic(PCR10) showed a range of 0.0134001 to 0.0444002 with sample 4. It is not being the highest and sample 1012 being the lowest. There are significant differences in the values obtained 0.011. This shows that PCR segment decreases with increase in working period. No individual segment that is up to PCR 104 (0.0134001).

445: 11'Spline(12,17,5,5)Asymptotic(PCR10)

The result obtained for 11'Spline(12,17,5,5)Asymptotic(PCR10) showed a range of 0.0134001 to 0.0444002 with sample 4. It is not being the highest and sample 1012 being the lowest. There are significant differences in the values obtained 0.011. This shows that PCR segment decreases with increase in working period, and they may not be connected with the different working material that was used. No individual segment that is up to PCR 104 (0.0134001).

44.11 *Sphagnum*(11,11,11,11) *subulna* (PCB 10)

The results obtained for *L1 Sphagnum*(11,11,11,11) range from 532.4000 to 63700000 with sample 10 being the highest value and sample 1117 being the lowest. There are no significant differences in the values obtained ($p > 0.05$). The calculated maximum load is set for PCB 10 (1454, 2003).

44.12 *Sphagnum*(11,11,11,11) *subulna* (PCB 10)

The results obtained for *L1 Sphagnum*(11,11,11,11) *subulna*(10 to 117) showed a range of 532.4000 to 63700000 sample 10 being the highest and sample 1117 being the lowest. There are no significant differences in the values obtained ($p > 0.05$). The shows that PCB 10 continues to decrease with increase in cutting period. The calculated maximum load is set for PCB 10.

44.13 *Deschmiedia*(10) (PCB 10)

PCB 10 is a polychlorinated biphenyl compound, containing six chlorine atoms on a biphenyl structure. It is extremely persistent, nonbiodegradable and tends to bioaccumulate in the environment and living organisms. PCB 10 can cause liver damage, endocrine disruption and neurological issues.

The results obtained for PCB 10 showed a range of 532.4000 to 63700000 sample 10 being the highest and sample 1117 being the lowest. There are no significant differences in the values obtained ($p > 0.05$). The shows that PCB 10 continues to decrease with increase in cutting period. The calculated maximum load is set for PCB 10.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

5.1 Conclusion

All previous ongoing technical issues with formation and deposits of F&G and H&B and human health concerns are investigated. Daily consumption of hot water and shower water should be less than 100 litres per person per day. The allowable limit which are given below to human health.

Exclusively, the length of working time is positive impact on the reduction of the consumption of the F&G. Therefore, all consumers of that time and time should consume around 100 litres of water if the possible is better and it is affected.

5.2 Recommendations

Engel rate from 10% to 20% and 20% to 30% are still to be consumed but should not be consumed every day. The rate of water must should be reduced and water regularly to prevent from any impact from entering the body.

The rate can be to control to reduce the consumption of the water. Consumer should reduce to half the water used before consumption. All consumers of that rate can take should adhere to only for a maximum 100 litres of the possible is better and it is affected.

北京德恒律师事务所

- [illegible]

European Commission (2004) Commission Regulation (EC) No 1831/2003 of 22 September 2004 setting maximum levels for certain substances in foodstuffs with maximums effective 1 January 2005 regarding food additives (EC No) (Official Journal of the European Union L24).

European Food Safety Authority (2014) (2015) *Physically-chemical hydrocolloids in food*. Scientific opinion of the Panel on Food Additives in the Food Chain. EFSA Journal 14(1).

Influence of poly(vinyl acetate) hydrocolloids on the in vitro digestibility of food. *Food J Sci Technol* 26 (Munro, S.S., H. Loh and D.J. Wedgley 2012).

FAO/WHO (2004) *Summary and conclusions*. Joint FAO/WHO expert consultation on food additives (JECFA) 64th meeting. Rome, 24-28 October 2003.

Food as a source of physico-chemical ingredients. *Food Science Food* 117:113-140. 2014. *Food J Sci Technol*, 27. Munro, S.S., M. Moham, M.S. Abdel-Hamid, S.B. El-Hachimi, S.I. El-Farouk and J. Mubarek 2015.

Global Environmental Distribution and Human Health Effects of Physically-chemical Hydrocolloids. In: *Global Environmental Trends of Potential Organic Chemicals*, Springer, 161, and P.R.S. Lee (Eds.) Chapter 7, Taylor & Francis Publication, Boca Raton, FL, USA. ISBN:1-4200-0444-0, pp. 47-74. 2010. Munro, S.S. and S.A. Quamra 2014.

Health risk assessment for Physically-chemical Hydrocolloids. #1144 project to develop composite of hydrocolloids as a Global scale. Physically-chemical Composite, www.chemsafety.org/chemsafety, 14. Munro, S.S., D.O. Elisei and S.I. El-Hachimi 2014.

Adeney, P. and Hutchins, A. Occupational demands and health problems among children in the Niger. *Niger Med J* 2010;55(1):109-112.

McKee, Oliver P., Davis, Michael P., MacHugh, David G., Johnston, I. 2010. "Unravelling the origin and biology of domestic animals using ancient DNA and phylogenetics" (<http://www.oxfordjournals.org/your/article/doi/10.1093/>)

Edmondson LR, Aggarwal NK, Fleming TR, Olson SE, Spiller CD, Heydari LR. Assessment of the public health importance of changing of diagnosis of gonorrhea using point-of-care rapid tests. *Journal of Infectious Diseases*. 2015;212:14-18.

Estimating and risk assessment of polycyclic aromatic hydrocarbons (PAHs) in processed food and feed raw materials. *Food Control* 42: 286-292. Li, Longji, L. H. Ouyang and Li, Enmei. 2017.

Christians, S., L. A. Aggarwal and P. Nataraj. 2010. Assessment of contamination of vegetable from urban and rural areas by heavy metals in India. *Appl. Biochem. Biotechnol* 2: 217-225.

Chen Y, Aggarwal NK, Jishi S, Shenghui B, Nataraj A. Determination of Food Metal Contamination in China (examined that students, Chinese. *Journal of Nutrition Research* 2014;4(1):111-114.

Estimating health risk assessment for and heavy metal contamination in the urban area of Harbin (China). *J. Soil & Water Res.* 49: 305-312-25. Wang Y, W. Qian Y, Li and J. Shi. 2012.

Proximate analysis, hydrocarbon determination in animal feed and chicken. *Polycyclic Aromatic Compounds* 36: 414-446-12. Murphy, B.L. and J. Shriver. 2001.

Appendix E GC with Chromatogram (H) of Hydrolyzed resin

Sample ID: 202304010001 (Hydrolyzed resin) Run Date: 2023-04-01
 Operator: J. Smith
 File Name: 202304010001_Hydrolyzed_resin.d File Size: 10.5 MB
 Run Time: 45:00:00
 Instrument Name: GC-MS
 Method Name: GC-MS (Hydrolyzed resin) Acquisition Mode: Scan Mode
 Column: DB-5MS (30m x 0.25mm ID) Carrier Gas: Helium
 Split Ratio: 1:10 Injection Volume: 1 µL
 Oven Temp: 40 °C (5 min) → 10 °C/min → 250 °C (10 min)
 Mass Spec: EI, 70 eV, Scan Range: 40-300 m/z
 Data File: 202304010001_Hydrolyzed_resin.d
 Report Generated: 2023-04-01 10:30:00



